

VIBRATIONS OF A CANTILEVER BEAM

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1 Abstract

In this lab, vibration analysis of a beam possessing distributed mass *and* end loading was performed by individually treating the vibration of (1), a massless beam with end loading, and (2), a beam with distributed mass without end loading, and superposing the results. Experimental values for Young's modulus E and beam density ρ were obtained through further post-processing of the vibration data. E was found to be 55.1938 GPa, while ρ was found to be 2740.0736 kg/m³. While it was hypothesized that for a relatively unchanging viscous damping coefficient d that the damping ratio ζ would increase with load, no clear-cut correlation was gleaned from the data.

2 Objectives

The primary objectives of this lab [1] were as follows:

1. To measure the natural frequencies ω_n and damping ratios ζ of a cantilever beam with various end loads.
2. To experimentally determine the modulus of elasticity E and density of the beam material ρ .

3 Introduction

3.1 Importance & Applications

Vibration analysis is a technique used to preemptively predict failure modes for structures like frames/trusses, mechanical componentry like shafts, and much more. In mechanical design, proper consideration must be given to the vibration of parts to ensure the fullest operational life and prevent failure under anticipated loadings.

3.2 Theoretical Basis

As paraphrased from [1]:

Given a (massless) rectangular beam of length L with some end load of mass M , Young's modulus E , and area moment of inertia I , its effective spring stiffness can be computed as

$$K = \frac{3EI}{L^3}, \quad (1)$$

with an undamped natural frequency

$$\omega_n = \sqrt{\frac{K}{M}}. \quad (2)$$

With the introduction of damping, the natural frequency changes to

$$\omega_d = \sqrt{1 - \zeta^2} \omega_n, \quad (3)$$

where ζ is the damping ratio (related to the viscous damping coefficient by $\zeta = d/(2\sqrt{KM})$).

For a distributed-mass beam with no end load, there exist infinite solutions corresponding to different vibrational modes. The lowest frequency mode of vibration occurs for $\lambda/L = 1.8751$ with

$$\lambda^4 = \frac{M_b \omega_2}{LEI}, \quad (4)$$

where M_b is the *actual* mass of the beam, ω_2 is the (undamped) natural frequency of the unloaded beam, and L, E, I are as above.

Both results can be superposed by using (4) to obtain an expression for ω_2 and equating it to (2), the expression for ω_n , yielding an expression for effective mass of the beam:

$$M_{\text{eff}} = 0.2427M_b. \quad (5)$$

3.3 Principal Findings

No explicit relationship was identified between the damping ratio ζ and the end mass M . However, the natural frequency ω_n was found to decrease with increasing end mass, as expected. The experimentally obtained Young's modulus E and density ρ were found to be 55.1938 GPa and 2740.0736 kg/m³, respectively, with percent uncertainties of 0.79% and 7.44%, respectively. Experimentally obtained values deviate slightly from those published for aluminum, which are 69 GPa for E and 2700 kg/m³ for ρ .

4 Procedures

4.1 Equipment

- Aluminum cantilever beam
- 2 C-clamps
- Mass balance
- 5 weights (≈ 100 g each)
- Weight hanger
- Proximity probe
- Instrumentation cables and connectors
- "Scope2023" VI
- Microsoft Excel (curve fit software)

4.2 Procedure

1. The mass of each of the 5 weights was measured using the mass balance and recorded. Beam dimensions were taken to be the same as measured in "Laboratory 4: Stress and Strain in a Cantilever Beam".
2. The aluminum cantilever beam and proximity probe were mounted to a rigid spot on the lab bench, above one of the front legs, using the C-clamps. The probe was connected to power and the oscilloscope with instrumentation cables; the preliminary signal was scaled on the oscilloscope.
3. The proximity probe signal was connected to the computer data acquisition system (channel 0 on the faceplate) using instrumentation cables. Upon manual excitation of the free end (using a finger), the "Scope2023" VI was used to capture the signal from the unloaded beam as the vibration decayed. The sample rate and number of sample was configured such that about 10 points per cycle were obtained and the vibration decayed to about 10% of the initial amplitude.
4. Step 3 was repeated for each load case, until the hanger and all 5 weights were attached to the free end of the beam.

5. Microsoft Excel was used to curve fit the time series data for each load case; the iterative refinement of the initial guesses for the best fit coefficients was performed using the “Solver” add-in.

5 Results

Figures 1-7 illustrate the vibrational response of the beam to excitation for each of the 7 load cases (unloaded, hanger only, hanger + 1 mass, hanger + 2 masses, hanger + 3 masses, hanger + 4 masses, hanger + 5 masses). In all figures, proximity probe output (in volts) is plotted against the elapsed time (in seconds). Conversion from voltages to displacements using the proximity probe calibration equation obtained in “Laboratory 2: Proximity Probe Calibration” was not undertaken since the main purpose of generating the graphs was to obtain frequency and damping ratio values for each load case (which can be derived from the best fit coefficients).

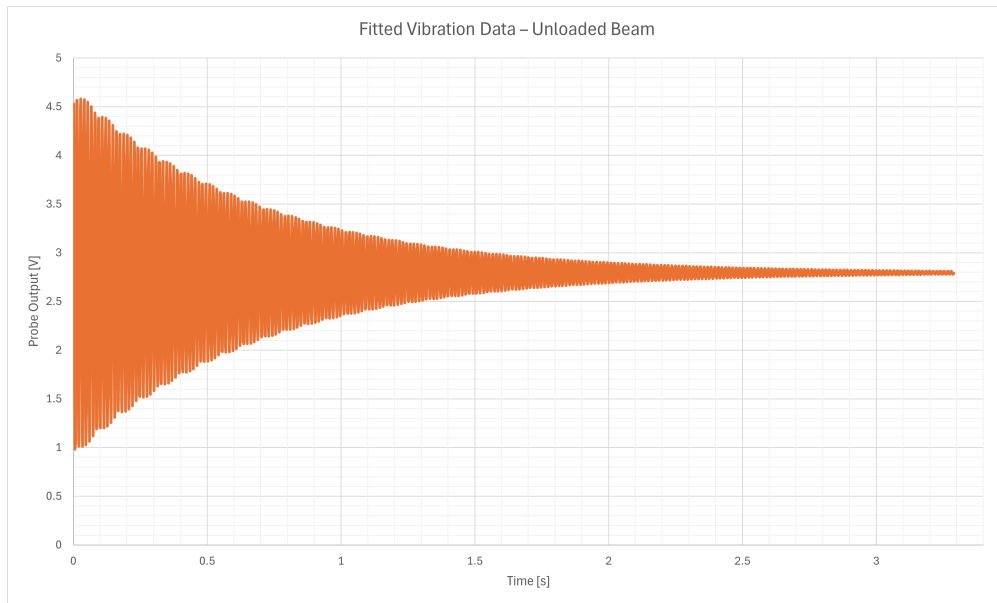


Figure 1: Curve-fitted vibration data for the unloaded beam.



Figure 2: Curve-fitted vibration data for the beam and hanger.

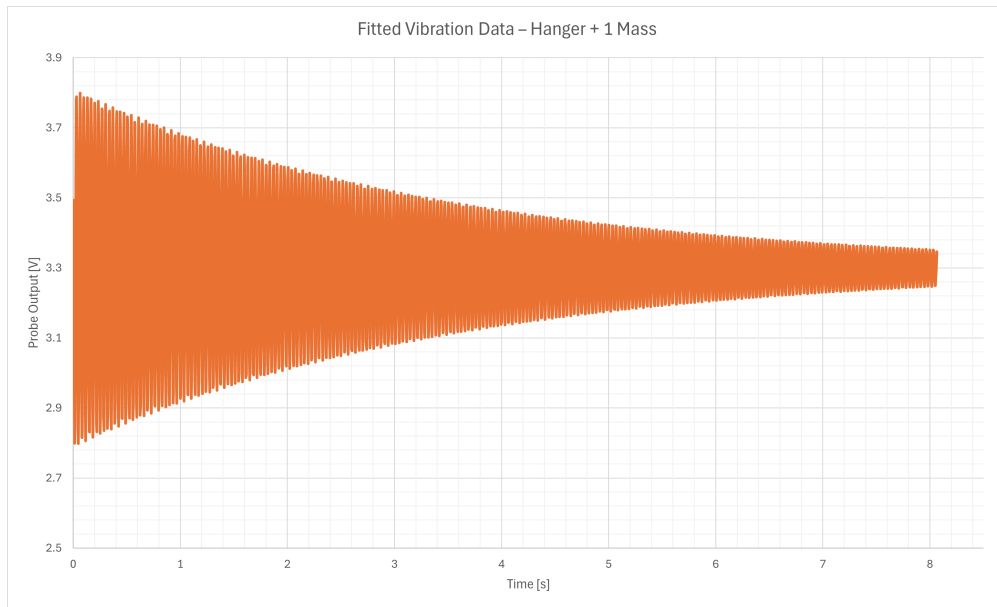


Figure 3: Curve-fitted vibration data for the beam, hanger, and 1 mass.

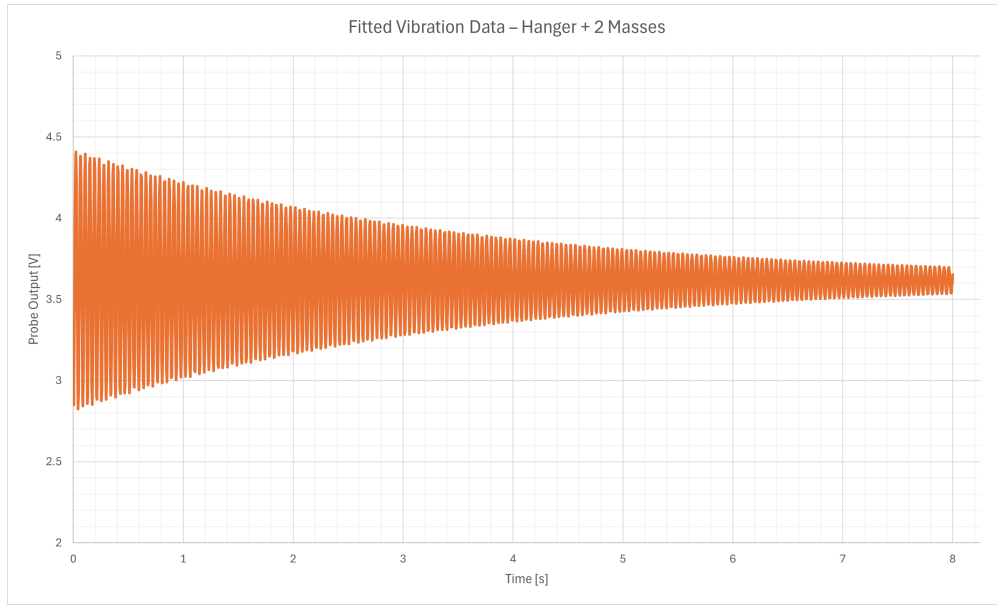


Figure 4: Curve-fitted vibration data for the beam, hanger, and 2 masses.

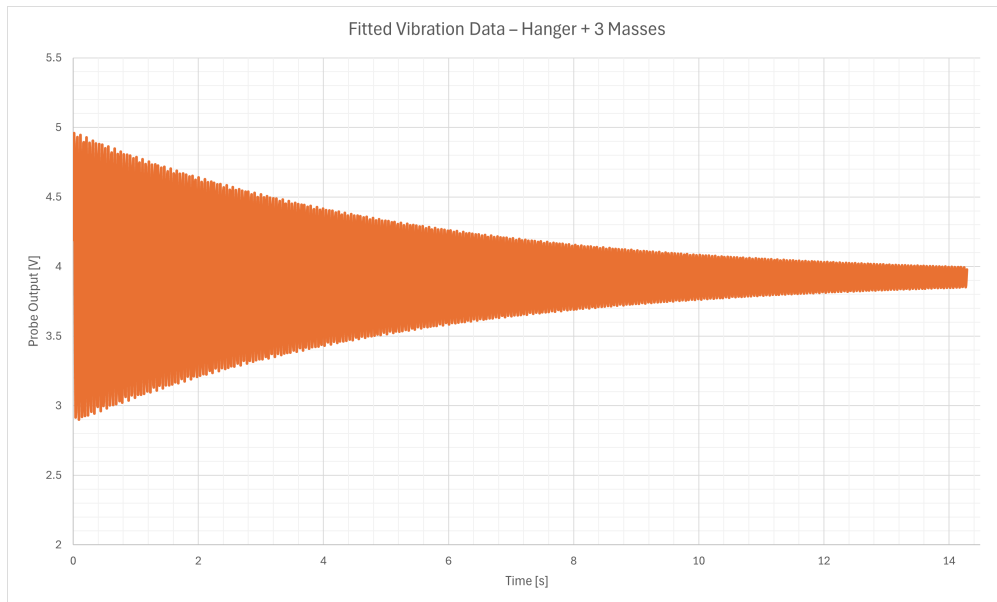


Figure 5: Curve-fitted vibration data for the beam, hanger, and 3 masses.

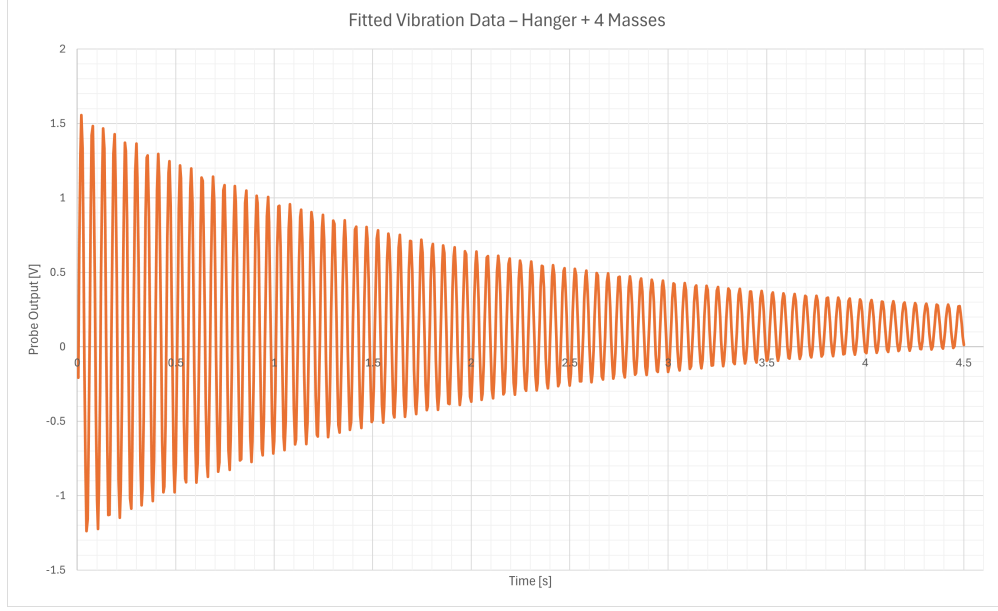


Figure 6: Curve-fitted vibration data for the beam, hanger, and 4 masses.

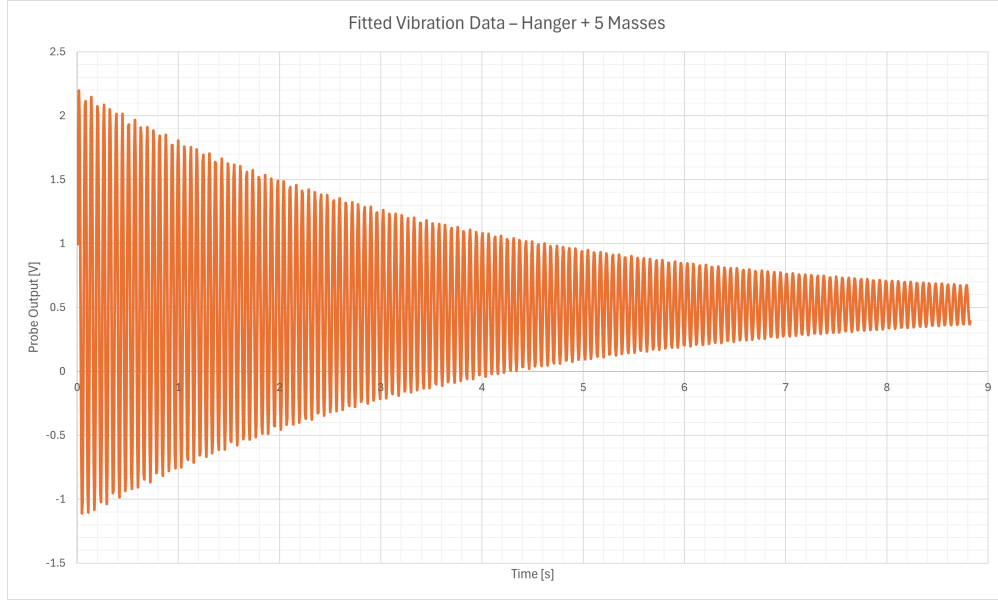


Figure 7: Curve-fitted vibration data for the beam, hanger, and 5 masses.

Consider the general form of the decaying sine wave equation for the damped vibration of a beam:

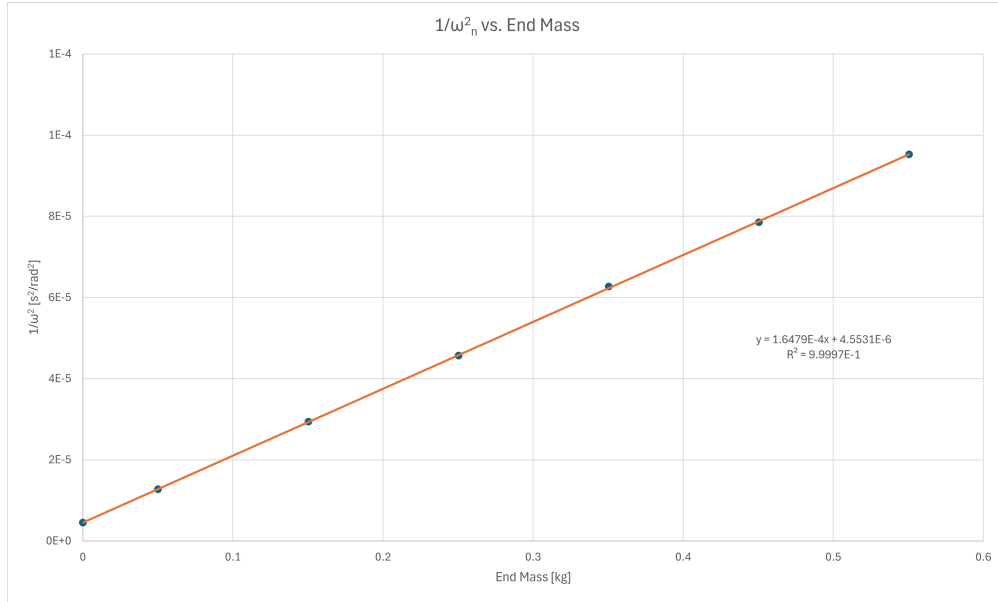
$$y = C_0 + C_1 e^{C_2 x} \sin(C_3 x + C_4), \quad (6)$$

For each load case, Microsoft Excel’s “Solver” add-in was used to optimize the best fit coefficients C_i such that the sum of squared errors was minimized. The resulting values for C_2 and C_3 were used to compute/obtain the natural frequencies ω_n and damping ratios ζ for each load case. The results are tabulated in Table 1.

Table 1: Dynamic response parameters for various beam load cases.

#	Load Case	End Mass M [kg]	C_2 [s] ⁻¹	Natural Freq. $C_3 = \omega_n$ [rad/s]	Damping Ratio ζ	$1/\omega_n^2$
1	Unloaded Beam	0	1.4466	469.386	0.003082	4.53879E-06
2	Beam + Hanger	0.05	0.4144	280.2045	0.001479	1.27365E-05
3	B + H + 1 Mass	0.1503	0.2828	184.5185	0.001533	2.93711E-05
4	B + H + 2 Masses	0.2504	0.2866	147.9464	0.001937	4.56868E-05
5	B + H + 3 Masses	0.3504	0.1874	126.3131	0.001483	6.26763E-05
6	B + H + 4 Masses	0.4504	0.5157	112.8209	0.004571	7.85635E-05
7	B + H + 5 Masses	0.5505	0.2724	102.4621	0.002659	9.52519E-05

Plotting the reciprocal squared natural frequencies $1/\omega_n^2$ against the end mass M for each load case, linear regression was performed to obtain experimental values for beam spring stiffness K (from the slope of the regression line) and effective beam mass M_{eff} (from the y-intercept). The plot and corresponding regression equation are shown in Figure 8 and (7), respectively.

**Figure 8:** Linear regression of $1/\omega_n^2$ against end mass M .

$$\frac{1}{\omega_n^2} = (1.6479\text{E-}4)M + 4.5531\text{E-}6 \quad (7)$$

From (7), K was found to be 6068.2935 N/m, while M_{eff} was found to be 0.02763 kg from the relation in (2). From (5), the actual beam mass M_b is computed to be 0.1138 kg. Now, these values, in conjunction with values for beam volume V and area moment of inertia I , can be used to find the material properties of the beam: namely, density ρ and Young's modulus E . Consider the beam properties tabulated below in Table 2:

Table 2: Beam properties.

Length L [m]	Width b [m]	Thickness h [m]
0.2508	0.02559	0.006473

Volume for the beam is computed as $L \times b \times h$, yielding $V = 4.1544\text{E-}5 \text{ m}^3$. Thus, the density ρ is computed as M_b/V , yielding $\rho = 2740.0736 \text{ kg/m}^3$. The area moment of inertia for the beam is computed as $I = bh^3/12$, yielding $I = 5.7832\text{E-}10 \text{ m}^4$. Finally, Young's modulus E can be computed by rearranging (1) to obtain $E = KL^3/(3I)$, yielding $E = 5.51938\text{E}10 \text{ Pa}$. In GPa, this is equivalent to $E = 55.1938 \text{ GPa}$. Refer to the Appendix for more detailed calculations. A summary of these results is tabulated in Table 3 below.

Table 3: Summary of experimentally obtained beam material properties.

Density ρ [kg/m ³]	Young's Modulus E [GPa]
2740.0736	55.1938

Uncertainty analysis yielded 7.44% error for ρ and 0.79% error for E . Detailed uncertainty calculations may be referenced in the Appendix.

6 Discussion & Conclusion

Analyzing Figure 8, the data points for $1/\omega_n^2$ versus end mass M appear to fall on a straight line, as expected from the theoretical relationship defined by (2), as the relationship stipulates that $(1/\omega_n^2)/M$ should produce a constant slope K . A high coefficient of determination of $R^2 = 0.99997$ indicates a near-perfect linear relationship.

Upon performing linear regression, the effective mass of the beam M_{eff} was found to be 0.02763 kg, which corresponds to an actual beam mass M_b of 0.1138 kg. The experimentally obtained Young's modulus E was found to be 55.1938 GPa, which is about 20% lower than the published value for aluminum (69 GPa). The experimentally obtained density ρ was found to be 2740.0736 kg/m³, which is about 1.5% higher than the published value for aluminum (2700 kg/m³). The discrepancies between experimental and published values for E and ρ may be attributed to a variety of factors, including measurement errors in beam dimensions, inaccuracies in the curve fitting process, and assumptions made in the theoretical model (e.g., neglecting modes of vibration corresponding to higher frequencies or damping effects).

Confidence intervals for E and ρ were found to be $E = 55.1938 \text{ GPa} \pm 0.4360 \text{ GPa}$ and $\rho = 2740.0736 \text{ kg/m}^3 \pm 203.8615 \text{ kg/m}^3$, respectively, at a 95% confidence level. The relatively low percent uncertainty for E (0.79%) indicates a high level of precision in the experimental determination of Young's modulus, while the higher percent uncertainty for ρ (7.44%) suggests that the density measurement is less precise, likely due to the cumulative effect of uncertainties in mass and volume measurements.

No clear-cut correlation was identified between the end mass M and the damping ratio ζ . While it was hypothesized that ζ would increase with M for a relatively unchanging viscous damping

coefficient d , the data does not appear to support this hypothesis. For instance, the damping ratio for the unloaded beam (0.003082) is higher than that of the beam with hanger + 1 mass (0.001533), which is counterintuitive since the latter has a higher end mass. The lack of a clear relationship may be attributed to experimental errors in curve fitting, as well as the fact that the damping ratio is influenced by multiple factors beyond just the end mass, such as material properties, surface roughness, and other sources of energy dissipation.

In conclusion, the vibration analysis of the cantilever beam yielded experimentally obtained values for Young's modulus and density that deviate from published values for aluminum, likely due to a combination of measurement errors and assumptions in the theoretical model. The relationship between end mass and damping ratio remains inconclusive, suggesting that further investigation is needed to fully understand the factors influencing damping in this system.

References

- [1] MECE 3360 Instructional Faculty, *SP2026 Lab Manual*, 2026, pp. 42-43.
- [2] Professor X. Chen, “Lecture 6 – Regression Analysis”, 2/18/2026
- [3] Professor X. Chen, “Vibration lab help file”, 4/22/2026

Appendix

Experimental Computation of Beam Density and Young’s Modulus

From (5) and Table 2:

$$\begin{aligned}\rho &= \frac{M_b}{V} = \frac{M_{\text{eff}}}{0.243V} = \frac{M_{\text{eff}}}{0.2427Lbh} = \frac{0.02763}{0.2427(0.2508 \cdot 0.02559 \cdot 0.006473)} \\ &= \boxed{2740.0736 \text{ kg/m}^3}\end{aligned}$$

From (1), Table 2, and K and I as computed in Results:

$$\begin{aligned}K &= \frac{3EI}{L^3} \Rightarrow E = \frac{KL^3}{3I} \\ E &= \frac{KL^3}{3I} = \frac{6068.2935 \cdot (0.2508)^3}{3 \cdot 5.7832\text{E-}10} \\ &= 5.51938\text{E}10 \text{ Pa} = \boxed{55.1938 \text{ GPa}}\end{aligned}$$

Uncertainty of Linear Regression Parameters

Note that (7) is of the form $y = cx + d$, where c is the slope and d is the y-intercept. The two-sided $1 - \alpha$ confidence intervals of these parameters can be computed using the following formulas, taken from [2]:

$$c \pm \frac{S_{y,x}t_{\frac{\alpha}{2},N-2}}{\sqrt{\sum_{i=1}^N(x_i^2 - N\bar{x})^2}} \quad (\text{i})$$

$$d \pm S_{y,x}t_{\frac{\alpha}{2},N-2}\sqrt{\frac{1}{N} + \frac{(\sum_{i=1}^N x_i)^2}{N^2(\sum_{i=1}^N x_i^2 - N\bar{x}^2)}} \quad (\text{ii})$$

where $S_{y,x}$ is the standard error of the estimate, $t_{\frac{\alpha}{2},N-2}$ is the critical value of the student’s t-distribution with $N - 2$ degrees of freedom, and N is the number of data points. The confidence intervals for c and d can be used to compute the uncertainty in K and M_{eff} , respectively, since these parameters are directly derived from the slope and y-intercept of the regression line. Referring to just the M and $1/\omega_n^2$ data points in Table 1 and leveraging Microsoft Excel functions (namely STEYX() and SUMSQ()), the following uncertainty parameters were found for an uncertainty $\alpha = 0.05$ (95% confidence level):

Standard Error of Estimate $S_{y,x}$	2.0616E-7
Critical Value $t_{\frac{\alpha}{2}, N-2}$	2.571
Number of Data Points N	7
Mean of M Data $\bar{M} \equiv \bar{x}$	0.2574

Substituting these values into (i) and (ii), the confidence intervals are found as below:

$$c \pm \frac{2.0616\text{E-}7 \cdot 2.571}{\sqrt{\sum_{i=1}^7 (M_i^2 - 7(0.2574)^2)^2}} \Rightarrow \boxed{(1.6479\text{E-}4 \pm 1.0546\text{E-}6) \text{ (N/m)}^{-1}}$$

$$d \pm 2.0616\text{E-}7 \cdot 2.571 \sqrt{\frac{1}{7} + \frac{(\sum_{i=1}^7 M_i)^2}{7^2(\sum_{i=1}^7 M_i^2 - 7(0.2574)^2)}} \Rightarrow \boxed{(4.5531\text{E-}6 \pm 3.3740\text{E-}7) \text{ kg} \cdot (\text{N/m})^{-1}}$$

Uncertainty of Young's Modulus and Density

Expressions for E and ρ can be written in terms of the slope c and y-intercept d by consulting the relations defined by (1) and (5). Note that in these expressions, the expressions for V and I , in terms of the beam dimensions, are substituted since their elemental uncertainties are known to be the resolutions of the instruments used to measure the dimensions. The expressions, as taken from [3], are as follows:

$$E = \frac{L^3}{3cI} = \frac{4L^3}{cbh^3} \quad (\text{iii})$$

$$\rho = \frac{M_b}{V} = \frac{M_{\text{eff}}}{0.2427V} = \frac{d}{0.2427cLbh} \quad (\text{iv})$$

Because (iii) and (iv) are both of the form $R = Cx_1^a x_2^b \dots x_n^N$, the percent error formula (v) can be used to compute the uncertainties of E and ρ . Uncertainty formulas for each are defined by (vi) and (vii), respectively:

$$\frac{w_R}{R} = \left[\left(a \frac{w_{x_1}}{x_1} \right)^2 + \left(b \frac{w_{x_2}}{x_2} \right)^2 + \dots + \left(N \frac{w_{x_n}}{x_n} \right)^2 \right]^{1/2} \quad (\text{v})$$

$$\frac{w_E}{E} = \left[\left(\frac{w_c}{c} \right)^2 + \left(3 \frac{w_L}{L} \right)^2 + \left(\frac{w_b}{b} \right)^2 + \left(3 \frac{w_h}{h} \right)^2 \right]^{1/2} \quad (\text{vi})$$

$$\frac{w_\rho}{\rho} = \left[\left(\frac{w_d}{d} \right)^2 + \left(\frac{w_c}{c} \right)^2 + \left(\frac{w_L}{L} \right)^2 + \left(\frac{w_b}{b} \right)^2 + \left(\frac{w_h}{h} \right)^2 \right]^{1/2} \quad (\text{vii})$$

w_c and w_d are defined as in the previous section. All other uncertainties are those arising from the measurement of the beam dimensions, which was performed using a digital caliper. From previous labs, the resolution of the caliper was found to be 0.01 mm, which is equivalent to 1E-5 m. Thus, $w_L = w_b = w_h = 1\text{E-}5$ m. For reference, all uncertainty values are tabulated below:

Referring to Table 2 for the beam dimensions, Table 3 for the experimentally obtained E and ρ

Uncertainty of Slope w_c	1.0546E-6 (N/m) ⁻¹
Uncertainty of Y-Intercept w_d	3.3740E-7 kg·(N/m) ⁻¹
Uncertainty of Length w_L	1E-5 m
Uncertainty of Width w_b	1E-5 m
Uncertainty of Thickness w_h	1E-5 m

values, the variable uncertainties above, and (7), percent uncertainties for E and ρ can be computed. Substituting all values into (vi) and (vii):

$$\frac{w_E}{E} = \left[\left(\frac{1.0546\text{E-}6}{1.6479\text{E-}4} \right)^2 + \left(3 \frac{1\text{E-}5}{0.2508} \right)^2 + \left(\frac{1\text{E-}5}{0.02559} \right)^2 + \left(3 \frac{1\text{E-}5}{0.006473} \right)^2 \right]^{1/2}$$

$$= 0.79\% \text{ or } w_E = 0.4360 \text{ GPa}$$

$$\frac{w_\rho}{\rho} = \left[\left(\frac{3.3740\text{E-}7}{4.5531\text{E-}6} \right)^2 + \left(\frac{1.0546\text{E-}6}{1.6479\text{E-}4} \right)^2 + \left(\frac{1\text{E-}5}{0.2508} \right)^2 + \left(\frac{1\text{E-}5}{0.02559} \right)^2 + \left(\frac{1\text{E-}5}{0.006473} \right)^2 \right]^{1/2}$$

$$= 7.44\% \text{ or } w_\rho = 203.8615 \text{ kg/m}^3$$

The final 95% confidence intervals for Young's modulus E and density ρ are given as follows:

$$\boxed{E = 55.1938 \text{ GPa} \pm 0.4360 \text{ GPa}}$$

$$\boxed{\rho = 2740.0736 \text{ kg/m}^3 \pm 203.8615 \text{ kg/m}^3}$$